

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

ANSWER KEY AND MARKING SCHEME

Biology: Class 9 (Science Group)

Syllabus: Chapter 1 (The Science of Biology) | Chapter 2 (Biodiversity) | Chapter 3 (The Cell) | Chapter 4 (Cell Cycle) | Chapter 5 (Tissues, Organs and Organ Systems) | Chapter 6 (Molecular Biology) | Chapter 7 (Metabolism, complete, with the main focus on 7.7 Photosynthesis and 7.8 Respiration)

Total Marks: 38

Time: 1 Hour 45 Minutes

Paper: Exam-17 (Version 1)

Answer letters: 1-C 2-A 3-D 4-C 5-B 6-A 7-B 8-D 9-A 10-B (A 3, B 3, C 2, D 2)

SECTION A: MCQ Key

#	Ans	Why (and why the others are wrong)	Source in her textbook
1	C chlorophyll of photosystem II	Chlorophyll a of photosystem II absorbs light and gives away a pair of electrons. Water splits to compensate this electron loss, which is why it is called photolysis of water.	Ch 7, 7.7.1.1 Light Dependent Reactions, steps 1 and 2 (p.103 to 104); SLO 7
2	A do not use light directly	The book says the second phase reactions do not use light directly, hence they are known as light independent reactions. They take place in the stroma, not in thylakoids, and they do not release oxygen.	Ch 7, 7.7.1 Mechanism of Photosynthesis (p.102) and 7.7.1.2 (p.104); SLO 7
3	D stomata	Step 3 of the light dependent reactions: oxygen is released out of the leaf through stomata. Cuticle is the waxy covering; xylem carries water and root hairs absorb it; none of them releases oxygen.	Ch 7, 7.7.1.1 Light Dependent Reactions, step 3 (p.104); SLO 7
4	C lack of light	Under the carton the grass got no light, so it could not photosynthesise. Air (oxygen and carbon dioxide) and soil water still reach the grass under a carton.	Ch 7, Exercise Section I, MCQ 20 (p.110); 7.7.2 light as a limiting factor; SLO 7
5	B 2 CO ₂ + 2 NADH	Book equation: 2 Pyruvic Acid + 2 CoA → 2 Acetyl CoA + 2 CO ₂ + 2 NADH. 4 CO ₂ and 2 FADH ₂ come from the Krebs cycle; 2 ATP + 2 NADH is glycolysis.	Ch 7, 7.8.3 Mechanism of Aerobic Respiration, 2. Formation of Acetyl CoA (p.106); SLO 8
6	A two ATP molecules	Book: electrons from each NADH give enough energy for three ATP, however each FADH ₂ produces only two ATP molecules.	Ch 7, 7.8.3, 4. Electron Transport Chain (p.107); SLO 8
7	B had no free oxygen in its atmosphere	Importance of anaerobic respiration, point 1: the earliest organisms respired anaerobically because the early earth had no free oxygen in its atmosphere.	Ch 7, 7.8.1 Importance of Anaerobic Respiration, point 1 (p.106); SLO 8
8	D wine making and baking	Yeast carries out alcoholic fermentation (ethyl alcohol + CO ₂), used in wine making and baking. Yogurt and cheese are made by bacteria; muscles make lactic acid.	Ch 7, 7.8.1 Importance of Anaerobic Respiration, points 2 and 3 (p.106); SLO 8
9	A brain	ATP usage in body: the brain uses maximum energy in the form of ATP, i.e. 25% of ATP.	Ch 7, 7.8.3 ATP usage in body, below Fig. 7.12 (p.107); SLO 6 and SLO 8
10	B water	Oxygen is the final electron acceptor and it ultimately combines with hydrogen ions to produce water. Carbon dioxide is released in acetyl CoA formation and the Krebs cycle.	Ch 7, 7.8.3, 4. Electron Transport Chain (p.107); SLO 8

SECTION B: Short Questions (model answers)

2(i) (4 marks)

(a) Tube **P** gives more bubbles. Limiting factor: **light intensity** (more light intensity and duration of light increase photosynthesis). **1**

(b) CO₂ is a raw material (substrate) of photosynthesis. More CO₂ concentration increases the rate, but **only up to a specific point**; after that, adding more does not increase it. **1**

(c) **25°C**, called the **optimum temperature** for photosynthesis. **1**

(d) The rate is **affected (lower)** than in P, because temperature below or above the optimum affects the rate. Accept "affected" or "lower". **1**

Source: Ch 7, 7.7.2 Limiting Factors of Photosynthesis (p.105); Exercise MCQ 17 pond weed idea (p.110); SLO 7.

2(ii) (4 marks)

(a) During tough, continuous (vigorous) exercise the **heart and lungs cannot provide enough oxygen** to the skeletal muscles for their increased requirement. **1**

(b) **Anaerobic respiration** (lactic acid fermentation). **1**

(c) Pyruvic Acid → Lactic Acid **1**

(d) Fig. 7.12 shows **2 ATP** from glycolysis and **36 ATP** from the electron transport chain. Both numbers needed for the mark. **1**
Source: Ch 7, 7.8.1 Lactic acid fermentation (p.105), glycolysis (p.105), Fig. 7.12 (p.107); SLO 8.

2(iii) (4 marks)**(a) RuBP, a 5-carbon molecule. 1****(b) A 6-carbon intermediate, which splits into two 3-carbon molecules (3PGA). 1****(c) NADPH and ATP** (3PGA is reduced by NADPH using the energy of ATP). **1****(d) G3P is used to synthesize glucose and to regenerate the original 5-carbon molecule (RuBP) for the next Calvin cycle. 1***Source: Ch 7, 7.7.1.2 Light Independent Reactions and Fig. 7.11 Calvin cycle (p.104 to 105); SLO 7.***SECTION C: Long Questions (model answers)****3(i) (8 marks)****(a) Diagram (3 marks):** a beaker half filled with water; Hydrilla branches at the bottom under an inverted glass funnel fully dipped in water; the funnel raised slightly on small supports; a water-filled test tube inverted over the funnel neck; gas collected at the top of the test tube and bubbles rising. Six labels: beaker, water + sodium bicarbonate, Hydrilla, funnel, test tube, gas (oxygen). **Neat drawing 1, six correct labels 2 (one mark for 3 to 5 labels).****(b)** It produces **CO₂ in the water** for the aquatic plant to use in photosynthesis. **1****(c)** For **mixing and free circulation of water. 1****(d)** Photosynthesis needs **light**; the set-up is kept in the sun or light for 2 to 3 hours so that photosynthesis occurs and gas collects. **1****(e)** The test tube is taken out and a **burning match stick** is put in; it **burns more brightly** ("more brightly" is needed for the mark). This confirms the gas is **oxygen** produced in photosynthesis. **1****(f) Photolysis of water** (splitting of water in the light dependent reactions). **1***Source: Ch 7, Activity 7.3 (p.102); 7.7.1.1 steps 2 and 3 (p.104); SLO 7.***3(ii) (8 marks)****(a) 1 + 1**Photosynthesis: $6\text{CO}_2 + 12\text{H}_2\text{O} \xrightarrow{\text{light, chlorophyll}} \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O} + 6\text{O}_2$ Aerobic respiration: $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{Energy (ATP)}$ **(b)** Any two, 1 mark each: **2**1. **Glucose** made in photosynthesis is the fuel broken down in respiration.2. **Oxygen** released in photosynthesis is used in aerobic respiration (final electron acceptor).3. **Carbon dioxide and water** produced in respiration are the raw materials of photosynthesis.**(c)** Four differences, 1 mark each: **4**

Feature	Photosynthesis	Aerobic respiration
Where	Chloroplasts (in plants)	Glycolysis in cytoplasm; Krebs cycle in mitochondrial matrix; ETC in inner mitochondrial membrane
Glucose	Made (from CO ₂ and water)	Broken down completely into CO ₂ and water
Energy	Light energy absorbed and stored in glucose	Energy released from glucose and stored as ATP
Type of metabolism	Anabolism	Catabolism
Gases	CO ₂ used, O ₂ released	O ₂ used, CO ₂ released

Source: Ch 7, 7.1 anabolism and catabolism examples (p.94); 7.7 equation (p.102); 7.8.2 aerobic equation (p.106); 7.8.3 sites (p.106 to 107); Exercise Section II Q14 and Section III Q9 (p.111); SLO 1, 7, 8.

End of Answer Key